



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

February 28, 2022

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 7945
(TWENTY-FIFTH REVISION)

EXPIRATION DATE: 2026-01-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Meggitt Safety Systems, Inc.
Simi Valley, CA
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale and use of a non-DOT specification cylinder conforming to all regulations applicable to DOT specification 4DS, except as specified herein, for the transportation in commerce of the materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a) party status may not be granted to a manufacturing permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.304a(a)(1) in that non-DOT specification cylinders are not authorized, except as specified herein.

5. BASIS: This special permit is based on Meggitt Safety Systems Inc.'s application dated February 3, 2022 and submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Compressed gas, n.o.s. Containing: (Bromotrifluoromethane, pentafluoroethane or iodotrifluoromethane, and nitrogen) OR: (NOVEC 1230(FK-5-1-12) and nitrogen) OR: (2-bromo-3,3,3-trifluoro-1-propene and carbon dioxide, and nitrogen)	2.2	UN1956	N/A
Fire Extinguishers	2.2	UN1044	N/A

7. SAFETY CONTROL MEASURES:

- a. PACKAGING: Packaging prescribed is a welded steel non-DOT specification cylinder conforming with DOT Specification 4DS (§§ 178.35 and 178.47) except as follows:

§ 178.47(a) *Type, size and service pressure.*

Applies except that service pressure must be at least 500 psig but not over 1,500 psig.

§ 178.47(b) *Steel.*

- (a) Type Nitronic 40 (ARMCO 21-6-9) austenitic stainless steel with analysis as follows:

Element	Percent
Carbon	0.04 Max.
Manganese	8.00/10.00
Phosphorus	0.060 Max.
Sulfur	0.030 Max.
Silicon	1.00 Max.
Chromium	19.00/21.50
Nickel	5.50/7.50
Nitrogen	0.15/0.40

§ 178.47(f) *Wall thickness.*

Minimum wall thickness for any cylinder shall be 0.040 inch. Minimum wall thickness shall be such that the wall stress does not exceed 45,000 pounds per square inch, at the marked service pressure.

- (1) Calculation for sphere must be made by the formula:

$$S = 0.25(PD/tE)$$

where:

S = Wall stress in pounds per square inch.

P = Service pressure, psig.

D = Outside diameter, inches.

t = Minimum wall thickness, inches.

E = 0.85 (provides 85 percent weld efficiency factor which must be applied in the girth weld area and heat affected zone which zone shall extend a distance of 6 times wall thickness from center of weld); E = 1.0 (for all other areas).

- (2) Calculation for a cylinder must be made by the formula:

$$S = P(1.3D^2 + 0.4d^2) / (D^2 - d^2)$$

where:

S = Wall stress in pounds per square inch.

P = Service pressure, psig.

D = Outside diameter, inches.

d = Inside diameter, inches.

§ 178.47(g) *Heat treatment.*

Cylinders may be stress relieved or annealed for forming. Welded cylinder shall be stress relieved at a temperature of $900 \text{ }^{\circ}\text{F} \pm 25 \text{ }^{\circ}\text{F}$ after process treatment and before hydrostatic test.

§ 178.47(n) *Acceptable results for flattening and burst tests.*

- (1) Flattening required to 50 percent of the original outside diameter without cracking except flattening test not required for spheres.
- (2) Burst pressure shall be at least 3 times service pressure or 4/3 times the test pressure whichever is greater.

b. MARKING: Markings must be in accordance with the requirements under § 178.35(g) except "DOT-SP 7945" must replace "DOT 4DS" followed by the cylinder service pressure, for example, DOT-SP 7945-XXX. The test Pressure, TP, must also be marked followed by the cylinder test pressure, for example, TP-XXXX. This marking must precede or be located immediately above the test date.

c. REQUALIFICATION: Each cylinder must be requalified in accordance with § 180.205, as prescribed for DOT Specification 4DS cylinders, except that the test pressure must be that stamped on the cylinder.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a packaging covered by this special permit, may reoffer it for transportation provided no modification

or change is made to the packaging and it is offered for transportation in conformance with this special permit

- c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
 - d. Each packaging manufactured under the authority of this special permit must be marked with a registration symbol designated by the Office of Hazardous Materials Safety (OHMS) for a specific manufacturing facility.
 - e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.
 - f. The cylinder must be shipped in strong outside packagings in accordance with § 173.301(a)(9).
 - g. The pressure vessels transported under this special permit may include properly approved actuating cartridges (Division 1.4C or 1.4S) installed in the discharge outlet without changing the classification of Division 2.2, provided the aggregate quantity of deflagrating (propellant) explosives does not exceed 3.2 grams per cylinder cartridge.
 - h. Cylinders are for Aircraft, Marine and Military Ground Vehicles use only.
 - i. Packages specified herein marked "DOT SP 5861" prior to March 20, 1976, or "DOT-E 5861" prior to June 1, 1978, and which meet the requirements of the edition of DOT SP 5861 or DOT-SP 5861 in effect at the time of manufacture, may be transported under the terms of this special permit.
 - j. A copy of the inspector's report specified in § 178.35(g) on the first lot of each design of cylinders produced with a service pressure greater than 900 psig must be submitted to the OHMSAPD prior to initial shipment.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, cargo aircraft only, and passenger-carrying aircraft.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel, aircraft or motor vehicle used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)– “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 Immediate notice of certain hazardous materials incidents, and 171.16 Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: ae